KERALA DIPLOMA CURRICULUM REVISION 2026

Course 5091 | 5 Credits: 4 | Program Core | Source-grounded

Tyre Technology

 To understand the development, classifications and nomenclature of pneumatic tyres.

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 5091 is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5091.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Polymer Technology
Course code	5091
Course title	Tyre Technology
Semester	5 Credits: 4
Course category	Program Core
Periods per week	4 (L:3, T:1, P:0) Periods per semester: 60

Item	Official information
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

16 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To understand the development, classifications and nomenclature of pneumatic tyres.
- To acquire knowledge on the different components of tyres, raw materials used and
- their design.
- To understand tyre building, curing and testing

Course outcomes

CO	Official outcome	Study level
CO1	15 Understanding pneumatic tyres. Summarize design and production of	See official syllabus
CO2	15 Understanding pneumatic tyre and tube Explain design and production of cycle tyre,	See official syllabus
CO3	14 Understanding cycle tube, solid tyre and re-treading	See official syllabus
CO4	Generalize tyre and tube testing methods 14 Understanding	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3
CO2	CO2 3 3
CO3	CO3 3 3
CO4	CO4 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Describe statistics, design and development of pneumatic tyres.	4	As specified
Module 2	Summarize design and production of pneumatic tyre and tube	4	As specified
Module 3	re-treading	4	As specified
Module 4	Generalize tyre and tube testing methods.	4	As specified

MODULE 1

Describe statistics, design and development of pneumatic tyres.

This module is presented from the official syllabus scope for Tyre Technology. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Describe statistics, design and development of pneumatic tyres.
- Official duration: As specified

- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

3 Remembering pneumatic tyres. Explain different components of tyre and

3 Remembering pneumatic tyres. Explain different components of tyre and is an official topic in Module 1 of Tyre Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Remembering pneumatic tyres. Explain different components of tyre and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 remembering pneumatic tyres. explain different components of tyre and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Remembering pneumatic tyres. Explain different components of tyre and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ 3 Remembering pneumatic tyres. Explain different components of tyre and □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

5 Understanding their functions

5 Understanding their functions is an official topic in Module 1 of Tyre Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Understanding their functions using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 understanding their functions should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Understanding their functions means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] 5 Understanding their functions [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Describe basic tyre constructions.

Describe basic tyre constructions. is an official topic in Module 1 of Tyre Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe basic tyre constructions. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe basic tyre constructions. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Describe basic tyre constructions. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] Describe basic tyre constructions. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Outline composition of tyre. 4 Understanding Content: History of pneumatic tyres. Statistics–production, consumption import and export of tyres. Major tyre manufacturing industries in India and abroad. Pneumatic tyres - definition - different components - functions. Different tyre construction - bias - radial - bias belted. Aspect ratio. Tyre size and dimension. Composition of tyre - elastomers - compounding ingredients (filler and chemicals) – tyre reinforcement. Requirements of tyre cord - different cords-cotton -rayon - nylon - polyester - glass - steel - aramid). Auxiliary materials

Outline composition of tyre. 4 Understanding Content: History of pneumatic tyres. Statistics–production, consumption import and export of tyres. Major tyre manufacturing industries in India and abroad. Pneumatic tyres - definition - different components - functions. Different tyre construction - bias - radial - bias belted. Aspect ratio. Tyre size and dimension. Composition of tyre - elastomers - compounding ingredients (filler and chemicals) – tyre reinforcement. Requirements of tyre cord - different cords-cotton -rayon -nylon - polyester - glass - steel - aramid). Auxiliary materials is an official topic in Module 1 of Tyre Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline composition of tyre. 4 Understanding Content: History of pneumatic tyres. Statistics–production, consumption import and export of tyres. Major tyre manufacturing industries in India and abroad. Pneumatic tyres - definition - different components - functions. Different tyre construction - bias - radial - bias belted. Aspect ratio. Tyre size and dimension. Composition of tyre - elastomers - compounding ingredients (filler and chemicals) – tyre reinforcement. Requirements of tyre cord - different cords-cotton -rayon -nylon - polyester - glass - steel - aramid). Auxiliary materials using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline composition of tyre. 4 understanding content: history of pneumatic tyres. statistics–production, consumption import and export of tyres. major tyre manufacturing industries in india and abroad. pneumatic tyres - definition - different components - functions. different tyre construction - bias - radial - bias belted. aspect ratio. tyre size and dimension. composition of tyre - elastomers - compounding ingredients (filler and chemicals) – tyre reinforcement. requirements of tyre cord - different cords-cotton -rayon -nylon - polyester - glass - steel - aramid). auxiliary materials should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline composition of tyre. 4 Understanding Content: History of pneumatic tyres. Statistics–production, consumption import and export of tyres. Major tyre manufacturing industries in India and abroad. Pneumatic tyres - definition - different components - functions. Different tyre construction - bias - radial - bias belted. Aspect ratio. Tyre size and dimension. Composition of tyre - elastomers - compounding ingredients (filler and chemicals) – tyre reinforcement. Requirements of tyre cord - different cords-cotton -rayon -nylon - polyester - glass - steel - aramid). Auxiliary materials means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Outline composition of tyre. 4 Understanding Content: History of pneumatic tyres. Statistics–production, consumption import and export of tyres. Major tyre manufacturing industries in India and abroad. Pneumatic tyres - definition - different components - functions. Different tyre construction - bias - radial - bias belted. Aspect ratio. Tyre size and dimension. Composition of tyre - elastomers - compounding ingredients (filler and chemicals) – tyre reinforcement. Requirements of tyre cord - different cords-cotton -rayon -nylon - polyester - glass - steel - aramid). Auxiliary materials □□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Describe statistics, design and development of pneumatic tyres.

              Describe history and development of
   M1.01                                                          3
Remembering
              pneumatic tyres.
              Explain different components of tyre and
   M1.02                                                          5
Understanding
              their functions
   M1.03      Describe basic tyre constructions.                  3
Understanding

   M1.04      Outline composition of tyre.                        4
Understanding

 Content:
 History of pneumatic tyres. Statistics–production, consumption import and export
of tyres.
 Major tyre manufacturing industries in India and abroad.
 Pneumatic tyres - definition - different components - functions. Different tyre
construction
- bias - radial - bias belted. Aspect ratio. Tyre size and dimension.
Composition of tyre - elastomers - compounding ingredients (filler and chemicals)
– tyre
reinforcement. Requirements of tyre cord - different cords-cotton -rayon -nylon -
polyester
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Summarize design and production of pneumatic tyre and tube

This module is presented from the official syllabus scope for Tyre Technology. Use the topic cards for learning and the source transcript for exact coverage.

As specified	**CO-linked**
official hours	see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Summarize design and production of pneumatic tyre and tube
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

6 Understanding components

6 Understanding components is an official topic in Module 2 of Tyre Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 6 Understanding components using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 6 understanding components should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 6 Understanding components means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] 6 Understanding components [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Compile adhesive treatments of tyre cords

Compile adhesive treatments of tyre cords is an official topic in Module 2 of Tyre Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Compile adhesive treatments of tyre cords using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, compile adhesive treatments of tyre cords should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Compile adhesive treatments of tyre cords means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] Compile adhesive treatments of tyre cords [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Summarize tyre building and curing 4 Understanding Outline design and production of automobile

Summarize tyre building and curing 4 Understanding Outline design and production of automobile is an official topic in Module 2 of Tyre Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize tyre building and curing 4 Understanding Outline design and production of automobile using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize tyre building and curing 4 understanding outline design and production of automobile should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarize tyre building and curing 4 Understanding Outline design and production of automobile means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] Summarize tyre building and curing 4 Understanding Outline design and production of automobile [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

3 Understanding inner tube and flap Content: Production of bead unit - inner liner - carcass - belt - breaker - tread - chafer - compound design for each component. Tread patterns - lug - rib - semi lug. Adhesive treatment for tyre cords- rayon - nylon -polyester. Adhesive formulation for each tyre cords. Tyre building machine - different parts - bias and radial building machines. Building process of bias and radial tyres. Curing of tyres - BOM and Auto form press. PCI - Flat spotting - Aqua planning. Curing bag - functions - properties - formulation. Automobile inner tube -production - typical compound. Tyre flap - production - typical compound. Explain design and production of cycle tyre, cycle tube, solid tyre and

3 Understanding inner tube and flap Content: Production of bead unit - inner liner - carcass - belt - breaker - tread - chafer - compound design for each component. Tread patterns - lug - rib - semi lug. Adhesive treatment for tyre cords- rayon - nylon -polyester. Adhesive formulation for each tyre cords. Tyre building machine - different parts - bias and radial building machines. Building process of bias and radial tyres. Curing of tyres - BOM and Auto form press. PCI - Flat spotting - Aqua planning. Curing bag - functions - properties - formulation. Automobile inner tube -production - typical compound. Tyre flap - production - typical compound. Explain design and production of cycle tyre, cycle tube, solid tyre and is an official topic in Module 2 of Tyre Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding inner tube and flap Content: Production of bead unit - inner liner - carcass - belt - breaker - tread - chafer - compound design for each component. Tread patterns - lug - rib - semi lug. Adhesive treatment for tyre cords- rayon - nylon -polyester. Adhesive formulation for each tyre cords. Tyre building machine - different parts - bias and radial building machines. Building process of bias and radial tyres. Curing of tyres - BOM and Auto form press. PCI - Flat spotting - Aqua planning. Curing bag - functions - properties - formulation. Automobile inner tube -production - typical compound. Tyre flap - production - typical compound. Explain design and production of cycle tyre, cycle tube, solid tyre and using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding inner tube and flap content: production of bead unit - inner liner - carcass - belt - breaker - tread - chafer - compound design for each component. tread patterns - lug - rib - semi lug. adhesive treatment for tyre cords- rayon - nylon -polyester. adhesive formulation for each tyre cords. tyre building machine - different parts - bias and radial building machines. building process of bias and radial tyres. curing of tyres - bom and auto form press. pci - flat spotting - aqua planning. curing bag - functions - properties - formulation. automobile inner tube -production - typical compound. tyre flap - production - typical compound. explain design and production of cycle tyre, cycle tube, solid tyre and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding inner tube and flap Content: Production of bead unit - inner liner - carcass - belt - breaker - tread - chafer - compound design for each component. Tread patterns - lug - rib - semi lug. Adhesive treatment for tyre cords- rayon - nylon -polyester. Adhesive formulation for each tyre cords. Tyre building machine - different parts - bias and radial building machines. Building process of bias and radial tyres. Curing of tyres - BOM and Auto form press. PCI - Flat spotting - Aqua planning. Curing bag - functions - properties - formulation. Automobile inner tube - production - typical compound. Tyre flap - production - typical compound. Explain design and production of cycle tyre, cycle tube, solid tyre and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 3 Understanding inner tube and flap Content: Production of bead unit - inner liner - carcass - belt - breaker - tread - chafer - compound design for each component. Tread patterns - lug - rib - semi lug. Adhesive treatment for tyre cords- rayon - nylon -polyester. Adhesive formulation for each tyre cords. Tyre building machine - different parts - bias and radial building machines. Building process of bias and radial tyres. Curing of tyres - BOM and Auto form press. PCI - Flat spotting - Aqua planning. Curing bag - functions - properties - formulation. Automobile inner tube -production - typical compound. Tyre flap - production - typical compound. Explain design and production of cycle tyre, cycle tube, solid tyre and □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Summarize design and production of pneumatic tyre and tube

             Comprehend design and production of tyre
  M2.01                                                                6
Understanding
             components
  M2.02      Compile adhesive treatments of tyre cords                 2
Understanding
  M2.03      Summarize tyre building and curing                        4
Understanding
             Outline design and production of automobile
  M2.04                                                                3
Understanding
             inner tube and flap
             Series Test I                                             1

Content:
Production of bead unit - inner liner - carcass - belt - breaker - tread - chafer
- compound
design for each component. Tread patterns - lug - rib - semi lug.
Adhesive treatment for tyre cords- rayon - nylon -polyester. Adhesive formulation
for
each tyre cords.
Tyre building machine - different parts - bias and radial building machines.
Building
process of bias and radial tyres. Curing of tyres - BOM and Auto form press. PCI
- Flat
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

re-treading

This module is presented from the official syllabus scope for Tyre Technology. Use the topic cards for learning and the source transcript for exact coverage.

As specified	**CO-linked**
official hours	see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: re-treading
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Outline design and production of cycle tyre

Outline design and production of cycle tyre is an official topic in Module 3 of Tyre Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline design and production of cycle tyre using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline design and production of cycle tyre should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline design and production of cycle tyre means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] Outline design and production of cycle tyre [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Describe design and production of cycle tube

Describe design and production of cycle tube is an official topic in Module 3 of Tyre Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe design and production of cycle tube using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe design and production of cycle tube should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Describe design and production of cycle tube means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Describe design and production of cycle tube □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Explain design and production of solid tyre 3 Understanding Describe design of re-treading materials and

Explain design and production of solid tyre 3 Understanding Describe design of re-treading materials and is an official topic in Module 3 of Tyre Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain design and production of solid tyre 3 Understanding Describe design of re-treading materials and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain design and production of solid tyre 3 understanding describe design of re-treading materials and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain design and production of solid tyre 3 Understanding Describe design of re-treading materials and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Explain design and production of solid tyre 3 Understanding Describe design of re-treading materials and □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

2 Understanding re-treading process Content: Cycle tyres: design - components production - building (monoband and collapsible drum method) - curing. Cycle tube: design and production. Solid tyres: formulations and production - applications - advantages and disadvantages - compare solid tyre and pneumatic tyre. Re-treading process: hot and cold process - machinery. compound designs (truck tyre tread, passenger car tread, rubber cement, cushion gum for conventional and pre-cured. Comparison of Hot and Cold re - treading.

2 Understanding re-treading process Content: Cycle tyres: design - components production - building (monoband and collapsible drum method) - curing. Cycle tube: design and production. Solid tyres: formulations and production - applications - advantages and disadvantages - compare solid tyre and pneumatic tyre. Re-treading process: hot and cold process - machinery. compound designs (truck tyre tread, passenger car tread, rubber cement, cushion gum for conventional and pre-cured. Comparison of Hot and Cold re - treading. is an official topic in Module 3 of Tyre Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding re-treading process Content: Cycle tyres: design - components production - building (monoband and collapsible drum method) - curing. Cycle tube: design and production. Solid tyres: formulations and production - applications - advantages and disadvantages - compare solid tyre and pneumatic tyre. Re-treading process: hot and cold process - machinery. compound designs (truck tyre tread, passenger car tread, rubber cement, cushion gum for conventional and pre-cured. Comparison of Hot and Cold re - treading. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding re-treading process content: cycle tyres: design - components production - building (monoband and collapsible drum method) - curing. cycle tube: design and production. solid tyres: formulations and production - applications - advantages and disadvantages - compare solid tyre and pneumatic tyre. re-treading process: hot and cold process - machinery. compound designs (truck tyre tread, passenger car tread, rubber cement, cushion gum for conventional and pre-cured. comparison of hot and cold re - treading. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding re-treading process Content: Cycle tyres: design - components production - building (monoband and collapsible drum method) - curing. Cycle tube: design and production. Solid tyres: formulations and production - applications - advantages and disadvantages - compare solid tyre and pneumatic tyre. Re-treading process: hot and cold process - machinery. compound designs (truck tyre tread, passenger car tread, rubber cement, cushion gum for conventional and pre-cured. Comparison of Hot and Cold re - treading. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 2 Understanding re-treading process Content: Cycle tyres: design - components production - building (monoband and collapsible drum method) - curing. Cycle tube: design and production. Solid tyres: formulations and production - applications - advantages and disadvantages - compare solid tyre and pneumatic tyre. Re-treading process: hot and cold process - machinery. compound designs (truck tyre tread, passenger car tread, rubber cement, cushion gum for conventional and pre-cured. Comparison of Hot and Cold re - treading. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
re-treading
  M3.01      Outline design and production of cycle tyre                5
Understanding
  M3.02      Describe design and production of cycle tube               4
Understanding
  M3.03      Explain design and production of solid tyre                3
Understanding
             Describe design of re-treading materials and
  M3.04                                                                 2
Understanding
             re-treading process
Content:
Cycle tyres: design - components production - building (monoband and collapsible
drum
method) - curing.
Cycle tube: design and production.
Solid tyres: formulations and production - applications - advantages and
disadvantages -
compare solid tyre and pneumatic tyre.
Re-treading process: hot and cold process - machinery. compound designs (truck tyre
tread, passenger car tread, rubber cement, cushion gum for conventional and pre-
cured.
Comparison of Hot and Cold re - treading.
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Generalize tyre and tube testing methods.

This module is presented from the official syllabus scope for Tyre Technology. Use the topic cards for learning and the source transcript for exact coverage.

As specified	CO-linked
official hours	see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Generalize tyre and tube testing methods.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

4 Understanding compounds Outline non-destructive and destructive

4 Understanding compounds Outline non-destructive and destructive is an official topic in Module 4 of Tyre Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding compounds Outline non-destructive and destructive using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding compounds outline non-destructive and destructive should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding compounds Outline non-destructive and destructive means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 4 Understanding compounds Outline non-destructive and destructive □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

4 Understanding testing of pneumatic tyres and tubes

4 Understanding testing of pneumatic tyres and tubes is an official topic in Module 4 of Tyre Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding testing of pneumatic tyres and tubes using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding testing of pneumatic tyres and tubes should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding testing of pneumatic tyres and tubes means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] 4 Understanding testing of pneumatic tyres and tubes [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Describe outdoor testing of pneumatic tyres

Describe outdoor testing of pneumatic tyres is an official topic in Module 4 of Tyre Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe outdoor testing of pneumatic tyres using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe outdoor testing of pneumatic tyres should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Describe outdoor testing of pneumatic tyres means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] Describe outdoor testing of pneumatic tyres [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Summarize testing of cycle tyre and tubes 4 Understanding Content: Testing of tyres - Nondestructive tests - dimension-size factor, tread arc width, non - skid depth, hardness, X-ray, IR, Ultra sonic, micro wave, holography. Destructive tests - endurance test and plunger test - cut tyre analysis. High way and Proving ground tests. Destructive and nondestructive tests of tyre inner tubes. Cycle tyre and tube tests - dimensions, tension set, cord strength, casing strength, Denison bend test, tensile and joint adhesion tests. Text /Reference: T/R BookTitle/Author Tyre Technology - F.J.Kovac (Publisher - Good Year Tyre & Rubber T1 Company) T2 Pneumatic Tyre Design - E. C. Woods (Publisher - W. Heffer) Rubber Technology and Manufacture - C.M.Blow R2 (Publisher - Butterworth Scientific) Vanderbilt Rubber Hand book - R.T Vanderbilt R3 (Publisher - Vanderbilt Company Inc.) R4 Rubber Technology and Manufacture - Published by SBP R5 IS Specifications - Bureau of Indian Standards Online resources: Sl.No Website Link 1 www.researchgate.net 2 www.scribd.com 3 www.iri.net.in

Summarize testing of cycle tyre and tubes 4 Understanding Content: Testing of tyres - Nondestructive tests - dimension-size factor, tread arc width, non - skid depth, hardness, X-ray, IR, Ultra sonic, micro wave, holography. Destructive tests - endurance test and plunger test - cut tyre analysis. High way and Proving ground tests. Destructive and nondestructive tests of tyre inner tubes. Cycle tyre and tube tests - dimensions, tension set, cord strength, casing strength, Denison bend test, tensile and joint adhesion tests. Text /Reference: T/R BookTitle/Author Tyre Technology - F.J.Kovac (Publisher - Good Year Tyre & Rubber T1 Company) T2 Pneumatic Tyre Design - E. C. Woods (Publisher - W. Heffer) Rubber Technology and Manufacture - C.M.Blow R2 (Publisher - Butterworth Scientific) Vanderbilt Rubber Hand book - R.T Vanderbilt R3 (Publisher - Vanderbilt Company Inc.) R4 Rubber Technology and Manufacture - Published by SBP R5 IS Specifications - Bureau of Indian Standards Online resources: Sl.No Website Link 1 www.researchgate.net 2 www.scribd.com 3 www.iri.net.in is an official topic in Module 4 of Tyre Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize testing of cycle tyre and tubes 4 Understanding Content: Testing of tyres - Nondestructive tests - dimension-size factor, tread arc width, non - skid depth, hardness, X-ray, IR, Ultra sonic, micro wave, holography. Destructive tests - endurance test and plunger test - cut tyre analysis. High way and Proving ground tests. Destructive and nondestructive tests of tyre inner tubes. Cycle tyre and tube tests - dimensions, tension set, cord strength, casing strength, Denison bend test, tensile and joint adhesion tests. Text /Reference: T/R BookTitle/Author Tyre Technology - F.J.Kovac (Publisher - Good Year Tyre & Rubber T1 Company) T2 Pneumatic Tyre Design - E. C. Woods (Publisher - W. Heffer) Rubber Technology and Manufacture - C.M.Blow R2 (Publisher - Butterworth Scientific) Vanderbilt Rubber Hand book - R.T Vanderbilt R3 (Publisher - Vanderbilt Company Inc.) R4 Rubber Technology and Manufacture - Published by SBP R5 IS Specifications - Bureau of Indian Standards Online resources: Sl.No Website Link 1 www.researchgate.net 2 www.scribd.com 3 www.iri.net.in using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize testing of cycle tyre and tubes 4 understanding content: testing of tyres - nondestructive tests - dimension-size factor, tread arc width, non - skid depth, hardness, x-ray, ir, ultra sonic, micro wave, holography. destructive tests - endurance test and plunger test - cut tyre analysis. high way and proving ground tests. destructive and nondestructive tests of tyre inner tubes. cycle tyre and tube tests - dimensions, tension set, cord strength, casing strength, denison bend test, tensile and joint adhesion tests. text /reference: t/r booktitle/author tyre technology - f.j.kovac (publisher - good year tyre & rubber t1 company) t2 pneumatic tyre design - e. c. woods (publisher - w. heffer) rubber technology and manufacture - c.m.blow r2 (publisher - butterworth scientific) vanderbilt rubber hand book - r.t vanderbilt r3 (publisher - vanderbilt company inc.) r4 rubber technology and manufacture - published by sbp r5 is specifications - bureau of indian standards online resources: sl.no website link 1 www.researchgate.net 2 www.scribd.com 3 www.iri.net.in should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarize testing of cycle tyre and tubes 4 Understanding Content: Testing of tyres - Nondestructive tests - dimension-size factor, tread arc width, non - skid depth, hardness, X-ray, IR, Ultra sonic, micro wave, holography. Destructive tests - endurance test and plunger test - cut tyre analysis. High way and Proving ground tests. Destructive and nondestructive tests of tyre inner

Aspect	What to prepare
	tubes. Cycle tyre and tube tests - dimensions, tension set, cord strength, casing strength, Denison bend test, tensile and joint adhesion tests. Text /Reference: T/R BookTitle/Author Tyre Technology - F.J.Kovac (Publisher - Good Year Tyre & Rubber T1 Company) T2 Pneumatic Tyre Design - E. C. Woods (Publisher - W. Heffer) Rubber Technology and Manufacture - C.M.Blow R2 (Publisher - Butterworth Scientific) Vanderbilt Rubber Hand book - R.T Vanderbilt R3 (Publisher - Vanderbilt Company Inc.) R4 Rubber Technology and Manufacture - Published by SBP R5 IS Specifications - Bureau of Indian Standards Online resources: Sl.No Website Link 1 www.researchgate.net 2 www.scribd.com 3 www.iri.net.in means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Summarize testing of cycle tyre and tubes 4 Understanding Content: Testing of tyres - Nondestructive tests - dimension-size factor, tread arc width, non - skid depth, hardness, X-ray, IR, Ultra sonic, micro wave, holography. Destructive tests - endurance test and plunger test - cut tyre analysis. High way and Proving ground tests. Destructive and nondestructive tests of tyre inner tubes. Cycle tyre and tube tests - dimensions, tension set, cord strength, casing strength, Denison bend test, tensile and joint adhesion tests. Text /Reference: T/R BookTitle/Author Tyre Technology - F.J.Kovac (Publisher - Good Year Tyre & Rubber T1 Company) T2 Pneumatic Tyre Design - E. C. Woods (Publisher - W. Heffer) Rubber Technology and Manufacture - C.M.Blow R2 (Publisher - Butterworth Scientific) Vanderbilt Rubber Hand book - R.T Vanderbilt R3 (Publisher - Vanderbilt Company Inc.) R4 Rubber Technology and Manufacture - Published by SBP R5 IS Specifications - Bureau of Indian Standards Online resources: Sl.No Website Link 1 www.researchgate.net 2 www.scribd.com 3 www.iri.net.in □□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Generalize tyre and tube testing methods.
              Describe testing on tyre cords and
  M4.01                                                              4
Understanding
              compounds
              Outline non-destructive and destructive
  M4.02                                                              4
Understanding
              testing of pneumatic tyres and tubes

  M4.03       Describe outdoor testing of pneumatic tyres            2
Understanding

  M4.03       Summarize testing of cycle tyre and tubes              4
Understanding


              Series Test II                                         1

 Content:
 Testing of tyres - Nondestructive tests - dimension-size factor, tread arc
width, non - skid
 depth, hardness, X-ray, IR, Ultra sonic, micro wave, holography. Destructive
tests -
 endurance test and plunger test - cut tyre analysis.
 High way and Proving ground tests.
 Destructive and nondestructive tests of tyre inner tubes. Cycle tyre and tube
tests -
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

> **Practice-paper notice:** The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain 3 Remembering pneumatic tyres. Explain different components of tyre and. (3 marks)

Answer: Begin with the standard definition or identity of *3 Remembering pneumatic tyres. Explain different components of tyre and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□□ □□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain 5 Understanding their functions. (3 marks)

Answer: Begin with the standard definition or identity of *5 Understanding their functions*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□□ □□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain Describe basic tyre constructions.. (3 marks)

Answer: Begin with the standard definition or identity of *Describe basic tyre constructions.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□□ □□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain Outline composition of tyre. 4 Understanding Content: History of pneumatic tyres. Statistics–production, consumption import and export of tyres. Major tyre manufacturing industries in India and abroad. Pneumatic tyres - definition - different components - functions. Different tyre construction - bias - radial - bias belted. Aspect ratio. Tyre size and dimension. Composition of tyre - elastomers - compounding ingredients (filler and chemicals) – tyre reinforcement. Requirements of tyre cord - different cords- cotton -rayon -nylon - polyester - glass - steel - aramid). Auxiliary materials. (3 marks)

Answer: Begin with the standard definition or identity of *Outline composition of tyre. 4 Understanding Content: History of pneumatic tyres. Statistics–production, consumption import and export of tyres. Major tyre manufacturing industries in India and abroad. Pneumatic tyres - definition - different components - functions. Different tyre* *construction - bias - radial - bias belted. Aspect ratio. Tyre size and dimension. Composition of tyre - elastomers - compounding ingredients (filler and chemicals) – tyre reinforcement. Requirements of tyre cord - different cords-cotton -rayon -nylon - polyester - glass - steel - aramid). Auxiliary materials*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 2

Define or explain 6 Understanding components. (3 marks)

Answer: Begin with the standard definition or identity of *6 Understanding components*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Compile adhesive treatments of tyre cords. (3 marks)

Answer: Begin with the standard definition or identity of *Compile adhesive treatments of tyre cords*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Summarize tyre building and curing 4 Understanding Outline design and production of automobile. (3 marks)

Answer: Begin with the standard definition or identity of *Summarize tyre building and curing 4 Understanding Outline design and production of automobile*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 3 Understanding inner tube and flap Content: Production of bead unit - inner liner - carcass - belt - breaker - tread - chafer - compound design for each component. Tread patterns - lug - rib - semi lug. Adhesive treatment for tyre cords- rayon - nylon -polyester. Adhesive formulation for each tyre cords. Tyre building machine - different parts - bias and radial building machines. Building process of bias and radial tyres. Curing of tyres - BOM and Auto form press. PCI - Flat spotting - Aqua planning. Curing bag - functions - properties - formulation. Automobile inner tube -production - typical compound. Tyre flap - production - typical compound. Explain design and production of cycle tyre, cycle tube, solid tyre and. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding inner tube and flap Content: Production of bead unit - inner liner - carcass - belt - breaker - tread - chafer - compound design for each component. Tread patterns - lug - rib - semi lug. Adhesive treatment for tyre cords- rayon - nylon -polyester. Adhesive formulation for each tyre cords. Tyre building machine - different parts - bias and radial building machines. Building process of bias and radial tyres. Curing of tyres - BOM and Auto form press. PCI - Flat spotting - Aqua planning. Curing bag - functions - properties - formulation. Automobile inner tube -production - typical compound. Tyre flap - production - typical compound. Explain design and production of cycle tyre, cycle tube, solid tyre and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 3

Define or explain Outline design and production of cycle tyre. (3 marks)

Answer: Begin with the standard definition or identity of *Outline design and production of cycle tyre*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain Describe design and production of cycle tube. (3 marks)

Answer: Begin with the standard definition or identity of *Describe design and production of cycle tube*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain Explain design and production of solid tyre 3 Understanding Describe design of re-treading materials and. (3 marks)

Answer: Begin with the standard definition or identity of *Explain design and production of solid tyre 3 Understanding Describe design of re-treading materials and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain 2 Understanding re-treading process Content: Cycle tyres: design - components production - building (monoband and collapsible drum method) - curing. Cycle tube: design and production. Solid tyres: formulations and production - applications - advantages and disadvantages - compare solid tyre and pneumatic tyre. Re-treading process: hot and cold process - machinery. compound designs (truck tyre tread, passenger car tread, rubber

cement, cushion gum for conventional and pre-cured. Comparison of Hot and Cold re - treading.. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding re-treading process Content: Cycle tyres: design - components production - building (monoband and collapsible drum method) - curing. Cycle tube: design and production. Solid tyres: formulations and production - applications - advantages and disadvantages - compare solid tyre and pneumatic tyre. Re-treading process: hot and cold process - machinery. compound designs (truck tyre tread, passenger car tread, rubber cement, cushion gum for conventional and pre-cured. Comparison of Hot and Cold re - treading.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 4

Define or explain 4 Understanding compounds Outline non-destructive and destructive. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding compounds Outline non-destructive and destructive*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 4 Understanding testing of pneumatic tyres and tubes. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding testing of pneumatic tyres and tubes*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Describe outdoor testing of pneumatic tyres. (3 marks)

Answer: Begin with the standard definition or identity of *Describe outdoor testing of pneumatic tyres*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Summarize testing of cycle tyre and tubes 4 Understanding Content: Testing of tyres - Nondestructive tests - dimension-size factor, tread arc width, non - skid depth, hardness, X-ray, IR, Ultra sonic, micro wave, holography. Destructive tests - endurance test and plunger test - cut tyre analysis. High way and Proving ground tests. Destructive and nondestructive tests of tyre inner tubes. Cycle tyre and tube tests - dimensions, tension set, cord strength, casing strength, Denison bend test, tensile and joint adhesion tests. Text /Reference: T/R BookTitle/Author Tyre Technology - F.J.Kovac (Publisher - Good Year Tyre & Rubber T1 Company) T2 Pneumatic Tyre Design - E. C. Woods (Publisher - W. Heffer) Rubber Technology and Manufacture - C.M.Blow R2 (Publisher - Butterworth Scientific) Vanderbilt Rubber Hand book - R.T Vanderbilt R3 (Publisher - Vanderbilt Company Inc.) R4 Rubber Technology and Manufacture - Published by SBP R5 IS Specifications - Bureau of Indian Standards Online resources: Sl.No Website Link 1 www.researchgate.net 2 www.scribd.com 3 www.iri.net.in. (3 marks)

Answer: Begin with the standard definition or identity of *Summarize testing of cycle tyre and tubes 4 Understanding Content: Testing of tyres - Nondestructive tests - dimension-size factor, tread arc width, non - skid depth, hardness, X-ray, IR, Ultra sonic, micro wave, holography. Destructive tests - endurance test and plunger test - cut tyre analysis. High way and Proving ground tests. Destructive and nondestructive tests of tyre inner tubes. Cycle tyre and tube tests - dimensions, tension set, cord strength, casing strength, Denison bend test, tensile and joint adhesion tests. Text /Reference: T/R BookTitle/Author Tyre Technology - F.J.Kovac (Publisher - Good Year Tyre & Rubber T1 Company) T2 Pneumatic Tyre Design - E. C. Woods (Publisher - W. Heffer) Rubber Technology and Manufacture - C.M.Blow R2 (Publisher - Butterworth Scientific)*

Vanderbilt Rubber Hand book - R.T Vanderbilt R3 (Publisher - Vanderbilt Company Inc.) R4 Rubber Technology and Manufacture - Published by SBP R5 IS Specifications - Bureau of Indian Standards Online resources: Sl.No Website Link 1 www.researchgate.net 2 www.scribd.com 3 www.iri.net.in. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **3 Remembering pneumatic tyres. Explain different components of tyre and**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **6 Understanding components**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Outline design and production of cycle tyre**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **4 Understanding compounds Outline non-destructive and destructive**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link www.researchgate.net www.scribd.com www.iri.net.in

IN-PAGE SEARCH

Search this lesson